BRIEF-001: Sovereign Weights: Quantized 405B Models in Air-Gap

Zero-Egress Enclaves | Air-Gapped Systems 2026

STAGE-X.SPACE TECHNICAL ARCHITECTURE BRIEF
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EXECUTIVE SUMMARY:
Field deployment report on operating quantized 405B-class models inside zero-egress enclaves. Describes weight partitioning, memory layout, and deterministic token generation without external dependencies or call-home verification.

KEY ARCHITECTURE PROOFS:
1. Weight Layout: FP8 quantization with block-wise scaling preserves 99.4% benchmark parity.
2. Memory Footprint: Fits on dual 8x H100 SXM nodes within local unified memory.
3. Kernel Egress: Enforced via netfilter drop_all at Linux kernel boundary [0 bytes leaked].
4. Verification: All inference outputs passed through strict confidence interlock gate.

FIELD MEASUREMENTS:
- P99 Cold Start TTFT: < 80ms on local weight storage
- Egress Packets: 0 packets outbound across 90-day soak test
- Requisition Time: Reduced from 72 hours to < 4 minutes